

E-08-audit-DtS-pressure v1: audit D_t S obstruction, pressure Hessian and spectral gap

ZFL Lab

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MISSION: Audit whether $D_t S$ can provide temporal control over $D_t e_i$, closing obstruction from E-07. INPUT E-07: $D_t e_i \sim \frac{e_j^T (D_t S) e_i}{\lambda_i - \lambda_j} e_j$, $D_t S \supset -\nabla^2 p$, $\mathcal{C}_{PROVEN} = \{C_E^{weak}, C_{L^3}\}$ frozen.

FORBIDDEN: $D_t S$ controlled $\Rightarrow D_t e_i$ controlled without spectral gap. Pressure bound $\Rightarrow$ PROVEN closure. C_E^{strong} promotion. Hiding denominator $\lambda_i - \lambda_j \rightarrow 0$.

AUDIT:

1. Exact identity for $D_t S$:

From NS: $D_t u = -\nabla p + \nu \Delta u$, $\nabla \cdot u = 0$. $\nabla u = S + \Omega$. Take gradient: $D_t \nabla u + (\nabla u)^2 = -\nabla^2 p + \nu \Delta \nabla u$.

Symmetric part:

$$D_t S + (S^2 + \Omega^2)_{\text{sym}} + (S\Omega + \Omega S)_{\text{sym}} = -\nabla^2 p + \nu \Delta S.$$

More precisely, writing transport on left:

$$\boxed{D_t S = -(u \cdot \nabla) S - (S^2 + \Omega^2)_{\text{sym}} - (S\Omega + \Omega S)_{\text{sym}} - \text{Hess}(p) + \nu \Delta S + \text{correction}}.$$

With $\Delta p = -\text{tr}[(\nabla u)^2] = -\text{tr}(S^2 + \Omega^2)$.

Thus $D_t S$ contains 5 families: (a) $-(u \cdot \nabla) S$ transport, (b) $-S^2$ quadratic strain self-amplification, (c) $-\Omega^2$ vorticity contribution, (d) $-\nabla^2 p$ pressure Hessian (nonlocal, $p = (-\Delta)^{-1} \text{tr}[(\nabla u)^2]$), (e) $\nu \Delta S$ dissipation.

2. Control analysis:

(a) $(u \cdot \nabla) S$: needs $\|u\|_\infty$, C_{L^3} gives $u \in L^3$ only, OPEN. (b) S^2 : $|S^2| \sim |S|^2$, quadratic, no L_t^1 bound from PROVEN. (c) Ω^2 : $|\Omega| \sim |\omega|$, unbounded if enstrophy diverges. (d) $\nabla^2 p$: Calderon-Zygmund $\|\nabla^2 p\|_{3/2} \lesssim \|\nabla u\|_3^2$, but $S \notin L^3$ from C_{L^3} alone. Nonlocal OPEN. (e) $\nu \Delta S$: dissipative but no pointwise bound.

Thus $\|D_t S\|$ not controlled by PROVEN.

3. Spectral gap critical:

Even if $|D_t S| \leq M(t)$ pointwise,

$$|D_t e_i| \leq \sum_{j \neq i} \frac{|e_j^T (D_t S) e_i|}{|\lambda_i - \lambda_j|}.$$

Need

$$\boxed{|D_t S| \leq M \text{ and } |\lambda_i - \lambda_j| \geq \delta > 0}$$

to get $|D_t e_i| \leq M/\delta$.

Gap $\delta \rightarrow 0$ corresponds to eigenvalue crossing, eigenvectors not differentiable, frame singular.
 No result in PROVEN prevents $\lambda_i - \lambda_j \rightarrow 0$ near T_* . Isotropic S gives small gap.

Hence even with pressure controlled, second obstruction: spectral gap.

4. Classification:

Exact identity for $D_t S$ with 5 terms, each requiring stronger norms than C_{L^3} or C_E^{weak} . Pressure Hessian is nonlocal and quadratic in u . Spectral gap not guaranteed.

OUTPUT:

$D_t S$ structure = $-(u \cdot \nabla)S - S^2 - \Omega^2 - \nabla^2 p + \nu \Delta S + \text{comm}$, EXACT

$\|D_t S\|$ not controlled by $\mathcal{C}_{PROVEN} = \{C_E^{weak}, C_{L^3}\}$

Even if $\|D_t S\| \leq M$, $|D_t e_i| \leq M/|\lambda_i - \lambda_j|$ blows up when gap $\rightarrow 0$

Pressure Hessian $\nabla^2 p = \text{Riesz of } u \otimes u$, needs L^∞ or stronger than L^3

Result:

PARTIAL/OPEN for $D_t S$ audit: exact decomposition obtained, but control OPEN.

BLOCKED for closing $D_t e_i$ via $D_t S$: two independent obstructions: (i) $\nabla^2 p$ nonlocal, not in L_t^1 from PROVEN, (ii) spectral gap $|\lambda_i - \lambda_j| \geq \delta$ not guaranteed.

Thus pathway spectral alignment -i control of W_- formally blocked at identifiable point: $D_t e_i$ due to pressure Hessian + eigenvalue crossing.

$\mathcal{C}_{PROVEN}$ remains $\{C_E^{weak}, C_{L^3}\}$, G locked at 0a37811.

Next: freeze E-07/E-08 as closed audit of alignment pathway.